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### HEADSET AND CONTROL METHOD WITH USAGE STATUS DETECTION

#### Abstract

A headset for audio play, comprising one or more sensor circuit and a control module. The control module comprises a secondary processor circuit. The sensor circuit generates sensor data according to structural deformations of the headset. The secondary processor circuit receives the sensor data and calculates measured values related to the structural deformations of the headset. The secondary processor circuit determines a usage status of the headset according to the measured value and one or more threshold value. The secondary processor circuit adaptively adjusts at least one threshold value according to the usage status.

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## Background/Summary

CROSS REFERENCE TO RELATED APPLICATION [0001] This application is a Continuation application of U.S. patent application Ser. No. 18/124,761, filed on Mar. 22, 2023, which claims the priority benefit of Chinese Patent Application Serial Number 202210286229.0, filed on Mar. 23, 2022. These and all other referenced extrinsic materials are incorporated herein by reference in their entirety.

### BACKGROUND

#### Technical Field

[0002] The application relates to a headset, in particular, to a headset control method which determines various usage statuses to perform various functions.

#### Related Art

[0003] Conventional headsets often detect a usage status indicating how the headset is used, so as to accordingly perform functions such as playing music or stop playing, and to achieve benefits such as power efficiency. At present, one of the most common way is the utilization of an infrared sensor. However, infrared sensors are often prone to false detection, and are insensitive to various statuses other than wearing and non-wearing.

### SUMMARY

[0004] In one aspect of application, an embodiment of a headset is provided for audio play, comprising at least a sensor circuit, a control module. The control module comprises a secondary processor circuit. The sensor circuit is arranged to operably generate sensor data based on structural deformations of the headset. The secondary processor circuit is arranged to operably receive the sensor data and calculate a measured value from the structural deformations of the headset. The secondary processor circuit is arranged to operably determine a usage status of the headset based on the measured value and one or more threshold value. The secondary processor circuit is arranged to adaptively adjust at least one of the one or more of threshold values based on the usage status.

[0005] In a further embodiment, when the secondary processor circuit determines from the usage status that the headset requires a primary processor circuit to function, the secondary processor circuit issues an event notification to the primary processor circuit, causing the primary processor circuit to be awoken from a sleep mode and to control the headset in response to the usage status.

[0006] In a further embodiment, when the measured value is lower than a first threshold value, the secondary processor circuit determines that the usage status is a non-wearing state. When the measured value is greater than or equal to the first threshold value, and lower than a second threshold value, the secondary processor circuit determines that the usage status is an open state. When the measured value is greater than or equal to the second threshold value, and lower than a third threshold value, the secondary processor circuit determines that the usage status is a wearing

state. When the measured value is greater than or equal to the third threshold value, the secondary processor circuit determines that the usage status is a stretch state.

[0007] In a further embodiment, the secondary processor circuit is arranged to operably monitor the measured value. When the usage status is the non-wearing state, and the secondary processor circuit detects that the measured value is increased to a first value greater than the first threshold value and maintains substantially around the first value for a time period, the secondary processor circuit determines that the usage status is switched to the wearing state, and configures the second threshold value to be the first value subtracting a predetermined offset.

[0008] In a further embodiment, the secondary processor circuit configures the third threshold value to be the second threshold value plus a fixed number. When the headset operates in the wearing state, and the secondary processor circuit detects that the measured value is increased to exceed the third threshold value, the secondary processor circuit determines that the usage status is the stretch state.

[0009] In another embodiment, when the headset is in the wearing state, and the secondary processor circuit detects that the measured value is increased to exceed a predetermined value within a default time, the secondary processor circuit determines that the usage status is in the stretch state.

[0010] In a further embodiment, when the headset is in the wearing state, and the secondary processor circuit detects based on the measured value that the usage status is changed to the non-wearing state, the secondary processor circuit notifies the primary processor circuit to stop an audio play operation, and notifies the primary processor circuit to enter the sleep mode.

[0011] In a further embodiment, when the headset operates in the wearing state, and the secondary processor circuit detects based on the measured value that the usage status is changed to the stretch state, the secondary processor circuit notifies the primary processor circuit to stop the audio play operation, or notifies the primary processor circuit to mute a call.

[0012] In a further embodiment, when the secondary processor circuit receives the sensor data, the secondary processor circuit performs a jitter filtering on the sensor data, and cumulates or averages the jitter filtered sensor data collected within a first duration, to generate the measured value.

[0013] In a further embodiment, the headset further comprises a wireless communication circuit and a memory device. The wireless communication circuit is arranged to perform a wireless communication with a host device. The memory device is adaptable for storing data and at least one of the one or more threshold values. The secondary processor circuit utilizes the wireless communication circuit to receive from the host device configurations related to the at least one of the one or more threshold values, and to store the configurations in the memory device.

[0014] In a further embodiment, the headset further comprises a headband, coupled to the sensor circuit and the control module, arranged to be resilient and providing fixation of the headset. The sensor circuit detects deformations of the headband to generate the sensor data. The sensor data comprises voltage values.

[0015] Another aspect of the application proposes an embodiment of a headset control method, for controlling a headset comprising a sensor circuit and a control module, wherein the control module comprises a secondary processor circuit coupled to the sensor circuit. The sensor circuit is utilized to detect structural deformations of the headset to generate sensor data, to receive the sensor data, and to calculate a measured value from the structural deformations of the headset. The secondary processor circuit determines a usage status of the headset based on the measured value and one or more threshold value. The secondary processor circuit adaptively adjusts at least one of the one or more threshold values based on the usage status.

[0016] The embodiment of the present application highlights at least the following advantages. The sensor circuit in the headset can detect a variety of usage statuses, allowing a variety of control functions to be realized. Furthermore, one or more threshold values for determining the usage status can be dynamically adjusted to meet the needs of various users. Overall, the embodiments of

the application release the potential to serve various additional functions in the headset while improving the power efficiency.

[0017] It should be understood, however, that this summary may not contain all aspects and embodiments of the present invention, that this summary is not meant to be limiting or restrictive in any manner, and that the invention as disclosed herein will be understood by one of ordinary skill in the art to encompass obvious improvements and modifications thereto.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

[0018] The features of the exemplary embodiments believed to be novel and the elements and/or the steps characteristic of the exemplary embodiments are set forth with particularity in the appended claims. The Figures are for illustration purposes only and are not drawn to scale. The exemplary embodiments, both as to organization and method of operation, may best be understood by reference to the detailed description which follows taken in conjunction with the accompanying drawings in which:

[0019] FIG. 1 is a schematic diagram of the headset.

[0020] FIG. 2 is a structural diagram of the headset operating in the non-wearing state according to an embodiment of the application.

[0021] FIG. 3 is a structural diagram of the headset operating in the open state according to an embodiment of the application.

[0022] FIG. 4 is a structural diagram of the headset operating in the wearing state according to an embodiment of the application.

[0023] FIG. 5 is a structural diagram of the headset operating in the stretch state according to an embodiment of the application.

[0024] FIG. 6 is a schematic diagram of the relationship between the measured value and the structural features of the headset according to an embodiment of the application.

[0025] FIG. 7 is a flowchart of the headset control method according to a further embodiment of the application.

[0026] FIG. 8 is a flowchart of the headset control method according to a further embodiment of the application.

[0027] FIG. 9 is a flowchart of the headset control method according to a further embodiment of the application.

### DETAILED DESCRIPTION OF THE EMBODIMENTS

[0028] The present invention will now be described more fully hereinafter with reference to the accompanying drawings, in which exemplary embodiments of the invention are shown. This present invention may, however, be embodied in many different forms and should not be construed as limited to the embodiments set forth herein. Rather, these embodiments are provided so that this present invention will be thorough and complete, and will fully convey the scope of the present invention to those skilled in the art.

[0029] Certain terms are used throughout the description and following claims to refer to particular components. As one skilled in the art will appreciate, manufacturers may refer to a component by different names. This document does not intend to distinguish between components that differ in name but function. In the following description and in the claims, the terms “include/including” and “comprise/comprising” are used in an open-ended fashion, and thus should be interpreted as “including but not limited to”. “Substantial/substantially” means, within an acceptable error range, the person skilled in the art may solve the technical problem in a certain error range to achieve the basic technical effect.

[0030] The following description is of the best-contemplated mode of conducting the invention.

This description is made for the purpose of illustration of the general principles of the invention and should not be taken in a limiting sense. The scope of the invention is best determined by reference to the appended claims.

[0031] Moreover, the terms “include”, “contain”, and any variation thereof are intended to cover a non-exclusive inclusion. Therefore, a process, method, object, or device that comprises a series of elements not only include these elements, but also comprises other elements not specified expressly, or may include inherent elements of the process, method, object, or device. If no more limitations are made, an element limited by “include a/an . . .” does not exclude other same elements existing in the process, the method, the article, or the device which comprises the element.

[0032] In the following embodiment, the same reference numerals are used to refer to the same or similar elements throughout the invention.

[0033] FIG. 1 is a schematic diagram of the headset **100** architecture of the embodiment of the application, explaining the basic electronic functions contained in the headset **100** comprises at least the sensor circuit **130**, the first audio module **210**, the second audio module **220** and the control module **230**.

[0034] The sensor circuit **130** is configured to generate sensor data according to the structural deformations of the headset **100**. The headset **100** in the embodiment is designed to include an elastic or resilient mechanism, which can change in shape with the wearer's conditions. The sensor circuit **130** can be used to detect the structural deformations caused by a user's wearing action to generate electronic signals, so that components in the headset **100** may process subsequent functions accordingly.

[0035] The first audio module **210** comprises a first audio circuit **142** and an audio capture circuit **146**, while the second audio module **220** comprises a second audio circuit **144**. Since the headset **100** is usually a binaural structure providing stereo effects, the arrangements of the first audio module **210** and the second audio module **220** are respectively corresponded to the user's two ears. The audio capture circuit **146** can be a common microphone or an array of multiple microphones, while the first audio circuit **142** and the second audio circuit **144** can be a common loudspeaker or a sound field array composed of speakers in a variety of different frequency bands. The first audio module **210** with the audio capture circuit **146** is not limited to be worn in the left ear or right ear. In a further embodiment, the audio capture circuit **146** is not limited to be arranged in the first audio module **210**, but may also be implemented in the second audio module **220**.

[0036] The control module **230** comprises at least the secondary processor circuit **120**, the primary processor circuit **110**, the memory device **140**, and the wireless communication circuit **150**. As an option, the control module **230** may also include a timer module **170** coupled to the primary processor circuit **110** and the secondary processor circuit **120**. The memory device **140** and the wireless communication circuit **150** are coupled to the primary processor circuit **110**. The secondary processor circuit **120** is coupled to the sensor circuit **130** and the primary processor circuit **110**. The primary processor circuit **110** is also coupled to the first audio circuit **142**, the second audio circuit **144** and the audio capture circuit **146**.

[0037] As shown in FIG. 1, the control module **230** is coupled to the sensor circuit **130**, wherein the secondary processor circuit **120** may receive the sensor data generated by the sensor circuit **130** and calculate the measured value of the structural deformations of the headset **100**. The primary processor circuit **110** is configured to control an audio play operation performed in the first audio circuit **142** and the second audio circuit **144**, or audio capture performed by the audio capture circuit **146** in the headset **100**. For example, the primary processor circuit **110** may receive audio signals from other devices through the wireless communication circuit **150** to play through the first audio circuit **142** and the second audio circuit **144**. In another embodiment, the memory device **140** may store audio files in advance, allowing the primary processor circuit **110** to read the audio files therefrom and play through the first audio circuit **142** and the second audio circuit **144**.

Furthermore, the audiofile stored in the memory device **140** may be generated by the primary processor circuit **110** controlling the audio capture circuit **146** to capture external sound. The audio file stored in the memory device **140** may also be received wirelessly from other devices through the wireless communication circuit **150** controlled by the primary processor circuit **110**.

[0038] In the embodiment, the primary processor circuit **110** and the secondary processor circuit **120** are arranged to coordinate with each other. The primary processor circuit **110** is a signal processor or microcontroller with powerful computation capability, capable of integrating and controlling all components in the headset **100**. The secondary processor circuit **120**, on the other hand, is implemented with a simpler chip of minor computation power than that of the primary processor circuit **110**, only needed to be capable of handling sensor data generated by the sensor circuit **130**. For example, when the headset **100** is less occupied, such as not performing an audio play operation, the primary processor circuit **110** may be switched to a sleep mode to save power. Meanwhile, the secondary processor circuit **120** keeps monitoring the sensor data of the sensor circuit **130**, and determines from the sensor data whether there is an event that needs to wake up the primary processor circuit **110** to function. Thus, when the secondary processor circuit **120** determines that there is a need to wake up the primary processor circuit **110**, the secondary processor circuit **120** may promptly send an event notification carrying information of a usage status describing how the headset **100** is used, so that the primary processor circuit **110** can be awakened seamlessly to perform corresponding functions according to the usage status. The event notification may also include further information such as the measured values, allowing the primary processor circuit **110** to determine more operation conditions according to the usage status combined with tracks of the measured values.

[0039] The following explains the operation mode of the headset **100** in terms of structural deformations.

[0040] FIG. 2 is a structural diagram of the headset **100** operating in non-wearing state according to the embodiment of the application. The headset **100** structurally comprises a headband **202**, the first earphone part **310** and the second earphone part **320**, in addition to the elements in FIG. 1. The headband **202** may be made with material or structure with elasticity or tension, wherein both ends are respectively connected with the first earphone part **310** and the second earphone part **320**. As an embodiment, the first audio module **210** and the control module **230** in FIG. 1 are arranged in the first earphone part **310**, while the second audio module **220** in FIG. 1 is arranged in the second earphone part **320**. When the headset **100** is worn on the head, the headband **202** stretches elastically, so that the user's ears are respectively contacted to the first earphone part **310** and the second earphone part **320** for the playback function. In the embodiment, the sensor circuit **130** is configured on the headband **202**, sensing the shape change of the headband **202** due to elastic stretching in use, so that sensor data is generated and sent to the secondary processor circuit **120** in the control module **230**. Taking FIG. 2 as an example, when the headset **100** is in the non-wearing state, the headband **202** is free from any stress or tension, and a non-wearing distance DN is naturally posed between the first earphone part **310** and the second earphone part **320**.

[0041] FIG. 3 is a structural diagram of the headset **100** operating in open state according to the embodiment of the application. In the embodiment, the open state is technically a special case in the non-wearing state, wherein the first earphone part **310** and the second earphone part **320** are subject to external forces to produce morphological changes, such as the open state distance DO. In this case, open state distance DO is subsequentially wider than non-wearing distance DN in FIG. 2, but not large enough to be worn on a regular user's head. Through proper configuration and calculation, the sensor circuit **130** may also be able to convert the tension or attitude change on the headband **202** into corresponding sensor data. In practical applications, the open state of the headset **100** may be used in a scenario wherein the user does not want to wear the headset **100** on his head, but still wants to put the headset **100** on the table as a general speaker. As long as the threshold values to figure out the usage status and corresponding workflows are designed, such a

use case desired by the user can be feasibly implemented in the headset **100**. The advantages of the embodiment are as follows. Unlike conventional headsets that can only be put on the head, a new application mode for the headset **100** is proposed. On the other hand, there is no need to design an added switch button to achieve the function needed in the embodiment. As long as the relative posture of the first earphone part **310** and the second earphone part **320** on the headband **202** is properly adjusted, the sensor circuit **130** is able to sense the condition and trigger successive operations. Therefore, the embodiment is cost-effective.

[0042] FIG. **4** is a structural diagram of the headset **100** operating in wearing state according to the embodiment of the application. The headset **100** in FIG. **4** has been worn on a user head **400**. The width of the user head **400** is wearing state distance DW. When wearing the headset **100**, an ordinary user must first apply a tensile force to the headband **202** of the headset **100** to make the distance between the first earphone part **310** and the second earphone part **320** greater than wearing state distance DW, and then place the headset **100** on the user head **400**. After placement to the user head **400**, the elasticity of the headband **202** causes the first earphone part **310** and the second earphone part **320** to be attached to the user head **400**, and finally converge stably to wearing state distance DW. It can be understood that in the above process, the sensor circuit **130** may detect significant variations of sensor data due to the condition of the headband **202** in a brief time, which eventually converge to a value corresponding to wearing state distance DW.

[0043] FIG. **5** is a structural diagram of the headset **100** operating in stretch state according to the embodiment of the application. When the headset **100** is worn to the user head **400** or removed from the user head **400**, it is often necessary to stretch the headband **202** to make the distance between the first earphone part **310** and the second earphone part **320** greater than wearing state distance DW, such as stretch state distance DS, so that it can be worn or removed. Therefore, determination of prying actions may help the secondary processor circuit **120** to find whether the user is putting on or off the headset **100**, so as to trigger corresponding functions. Furthermore, the action of prying away can also be used as extra command inputs. For example, while the headset **100** is worn on the user for music listening, if a pry way action is detected that causes the first earphone part **310** and the second earphone part **320** to be stretched above stretch state distance DS, the primary processor circuit **110** can be triggered to pause music or to perform other functions.

[0044] FIG. **6** is a schematic diagram of relationships between the measured value and the structural features of the headset **100** in the embodiment of the application. The structural deformations of the headband **202** in the earlier embodiment, that is, the relative distance between the first earphone part **310** and the second earphone part **320**, presents a specific linear relationship for the sensor circuit **130**. For example, when the distance between the first earphone part **310** and the second earphone part **320** is non-wearing distance DN, open state distance DO, wearing state distance DW, and stretch state distance DS, the sensor circuit **130** may obtain sensor data with corresponding non-wearing value QN, open state value QO, wearing state value QW, prying state value QS. In practical use, the sensor data received by the sensor circuit **130** may be presented in the form of voltage or current signals, or may be presented as digital signal. On the other hand, along a time axis (not shown), it can be understood that sensor data may carry jitter noises caused by user actions. Therefore, after the secondary processor circuit **120** of the embodiment receives sensed raw data from the sensor circuit **130**, further steps are needed to improve accuracy of determining the usage status. For example, the secondary processor circuit **120** continuously receives the sensor data from the sensor circuit **130** in time, then performs a jitter filtering on the sensor data to mitigate noises generated by the user's action, and performs a hysteresis operation to determine measured values that actually reflect the user actions. The hysteresis operation may comprise accumulating or averaging a plurality of the sensor data continuously collected for a period of time, so as to eliminate the influence caused by instantaneous error and to calculate a more accurate measured value. In a further embodiment, the headset **100** may define some threshold values in-between the measured values corresponding to the aforementioned distances,

such as first threshold value Q1, second threshold value Q2, and third threshold value Q3. By monitoring the measured value and comparing the measured value with the threshold values, the usage status of the headset **100** can be determined. For example, the measured values consecutively changed between multiple threshold values can be interpreted as one or more specific command inputs. For example, prying the headset **100** once, twice, or three times within a certain time may be defined as triggers of various functions.

[0045] FIG. 7 is a flowchart of a handset control method according to an embodiment of the present application. Based on the operation principle of the sensor circuit **130** in FIG. 6, it can be further developed into the embodiment of FIG. 7. To facilitate understanding, the steps of the sensor circuit **130**, the secondary processor circuit **120** and the primary processor circuit **110** are divided into three columns in FIG. 7.

[0046] Firstly, in step **702**, the tension or structural deformations on the headband **202** is sensed by the sensor circuit **130** to generate sensor data.

[0047] In step **704**, the secondary processor circuit **120** continuously receives sensor data from the sensor circuit **130**. Therefore, the form of the received sensor data on the time axis may be presented as a signal waveform similar to the form of voltage or current.

[0048] As described, although the sensor data corresponds to the usage status of the headset **100**, the sensor data still needs to be calculated and processed to ensure accurate determination. Therefore, in step **706**, the secondary processor circuit **120** performs jitter filtering on the received sensor data. The jitter filtering may involve a low pass filter, which cuts high frequency interferences in the user actions. On the other hand, to determine the usage status more accurately, the secondary processor circuit **120** may also employ a hysteresis operation to process sensor data. That is, aggregating, averaging, or statistically analyzing the sensor data collected from a consecutive period of time, to generate the measured value that reflects the usage status. For example, the measured value may be calculated from sensor data collected in three seconds. To achieve the hysteresis operation, the sensor data being collected are temporarily stored in the memory device **140** in step **708**. In step **710**, the timer module **170** is used to count down a hysteresis time limit. For example, if the elapsed time of sensor data collection does not reach three seconds, the process returns to step **704** to continue receiving sensor data. When the hysteresis time limit is reached, the secondary processor circuit **120** runs step **712** to generate measured value according to the sensor data stored in the memory device **140**. Understandably, the method of calculating measured value from sensor data may include summation, average, statistical analysis or a combination thereof. In practical application, the actual formula can be flexibly adjusted according to requirements for the sensitivity and accuracy of sensor data, not specifically limited in the embodiment.

[0049] In step **714**, when the secondary processor circuit **120** obtains the measured value standing for the current usage status, the secondary processor circuit **120** thereafter determines whether it is necessary to wake up the primary processor circuit **110** to process corresponding operations according to the measured value. If not, the process returns to step **704**, wherein the secondary processor circuit **120** keeps watching the sensor data. Conversely, if the secondary processor circuit **120** decides that the currently obtained measured value meets the conditions of particular events that need to be handed over to the primary processor circuit **110** for following operations, in step **716**, the secondary processor circuit **120** sends an event notification to the primary processor circuit **110**. When the usage status is changed, for example, from the open state to the wearing state, or from the wearing state to the stretch state, step **716** is processed.

[0050] On the other hand, the primary processor circuit **110** of the embodiment is designed to conditionally enter the sleep state. For example, the primary processor circuit **110** controls the headset **100** for some regular operations in the described step **718**, such as controlling the first audio module **210** and the second audio module **220** to play or capture audio, or controlling the wireless communication circuit **150** to make voice a call with a remote entity. In step **720**, when the



headset **100** no longer requires the operations and is less occupied, the primary processor circuit **110** is deemed satisfying a criteria to sleep, and therefore switched to the sleep mode in step **722**. [0051] Following step **716**, in step **724**, the primary processor circuit **110** in sleep is awakened after receiving the event notification, and the corresponding control of the headset **100** can be executed according to the information provided in the event notification, such as the current usage status and measured value. The primary processor circuit **110** may operate according to the usage status and measured value after being awakened, which can be, but is not limited to, playing music, making a call, answering a call, pausing playback, adjusting the volume, adjusting the sound effect and other device configuration, fast switching or playback, previous or next song, turning off the power, or triggering the mobile phone to run specific applications.

[0052] The embodiment of FIG. 7 illustrates the process of generating sensor data from the sensor circuit **130** and converting it to measured value according to the structural deformations of the headset **100**, and describes how the primary processor circuit **110** is matched with the secondary processor circuit **120** to sleep and save power. The present embodiment highlights at least the following advantages. The primary processor circuit **110** and the secondary processor circuit **120** are arranged to be two separate function units. Therefore, when tasks with high power consumption, such as music play or voice call, are no longer needed, the primary processor circuit **110** is allowed to sleep to save power consumption of the headset **100**. The secondary processor circuit **120** continuously performs monitor operations with low power consumption. Until an event requiring the primary processor circuit **110** occurs, the secondary processor circuit **120** wakes up the primary processor circuit **110**. Based on the operation principle of FIG. 7, the measured value calculated by the secondary processor circuit **120** is further compared with various threshold values to determine the change of usage status, or implement various forms of input instructions to derive diversified control methods. For example, the secondary processor circuit **120** can determine one of the usage statuses of the headset **100** according to the measured value and at least one threshold value, and wake up the primary processor circuit **110** to provide the functions required by the usage status. For example, multiple threshold values may be predetermined in the headset **100**, each corresponding to a different usage status. When the measured value is the lower than first threshold value **Q1**, the secondary processor circuit **120** determines that the usage status is non-wearing state. When the measured value is greater than or equal to the first threshold value **Q1** and lower than second threshold value **Q2**, the secondary processor circuit **120** determines that the usage status is the open state. When the measured value is greater than or equal to the second threshold value **Q2** and lower than third threshold value **Q3**, the secondary processor circuit **120** determines that the usage status is the wearing state. When the measured value is greater than or equal to the third threshold value **Q3**, the secondary processor circuit **120** determines that the usage status is the stretch state. In a further embodiment, the primary processor circuit **110** is also designed to adaptively adjust one of the threshold values according to the usage status and the measured value, as described in detail below.

[0053] FIG. 8 is a flowchart of the handset control method according to a further embodiment of the present application. In the embodiment, the detailed process of switching the headset **100** from the non-wearing state to the wearing state is described. First, in step **800**, the headset **100** operates in the non-wearing state. Meanwhile, the distance between the first earphone part **310** and the second earphone part **320** is shown as non-wearing distance **DN** in FIG. 2. In step **802**, the secondary processor circuit **120** determines whether the measured value calculated from the sensor data of the sensor circuit **130** is changed. It can be understood that although the process in FIG. 8 is not specifically described, the operations performed by the secondary processor circuit **120** and the sensor circuit **130** are resident in the background throughout each state, running steps as shown in steps **702** to **714** of FIG. 7.

[0054] When a user puts the headset **100** on the user head **400**, the sensor circuit **130** first senses a sudden data variation. As the headset **100** is settled on the user head **400**, the data sensed by the

sensor circuit **130** eventually stay on wearing state value QW corresponding to wearing state distance DW. Therefore, if the secondary processor circuit **120** detects that the measured value is changed in step **802**, such as increased to a first value greater than first threshold value Q1, in step **804**, the secondary processor circuit **120** further monitors whether the measured value can be maintained around the first value within a time limit, such as three seconds. If step **804** is also true, the secondary processor circuit **120** confirms that the usage status of the headset **100** has been switched to the wearing state.

[0055] On the other hand, when the headset **100** is changed from the non-wearing state to the open state, steps **802** and **804** are also applicable to determine whether to enter the wearing state.

[0056] After entering the wearing state, since the user head **400** width varies with different user, the sensor circuit **130** of the embodiment determines the wearing state value QW corresponding to the wearing state distance DW to adaptively adjust the threshold value of the wearing state. In step **806**, the secondary processor circuit **120** may update second threshold value Q2 according to the measured value given by sensor **130**. For example, second threshold value Q2 may be the wearing state value QW subtracting a predetermined offset, so that second threshold value Q2 is in a range lower than wearing state value QW but not less than the range of open state value QO. It is understandable that the predetermined offset can be flexibly determined according to the design requirements of the entity. After confirming the new second threshold value Q2, the headset **100** proceeds to step **810** to operate in the wearing state.

[0057] If the measured value does not remain stable in step **804**, step **808** is proceeded for further determination. In a further embodiment, the measured value change detected in step **802** may be one of a consecutive command inputs. For example, two or three consecutive prying actions can be defined as commands to run distinct functions. Therefore, in step **808** of FIG. 8, changes of the measured value are further tracked by the secondary processor circuit **120** to identify commands that may be input.

[0058] The second threshold value Q2 is configured differently for users with different head sizes, therefore the measured value changes differently when the headset **100** is pried. In step **810**, the headset **100** operates in the wearing state. The primary processor circuit **110** may set a new third threshold value Q3 according to the updated second threshold value Q2 in step **812**. For example, if second threshold value Q2 is the first value, third threshold value Q3 is a second value calculated from the first value adding a fixed number. It is understandable that the fixed number can be flexibly determined according to the requirements of practical implementations. In step **814**, the sensor circuit **130** and the secondary processor circuit **120** likewise, keeps monitoring the measured value. If the secondary processor circuit **120** detects that the measured value is increased to exceed third threshold value Q3 in step **816**, the secondary processor circuit **120** deems that the headset **100** is in a stretch state, so that step **818** is proceeded.

[0059] Determination of the stretch state may also be implemented in different approaches. For example, when the headset **100** operates in the wearing state in step **810**, and the secondary processor circuit **120** detects that the measured value increases by more than a predetermined value within a predetermined time value in step **814**, the secondary processor circuit **120** may instantly assert that the headset **100** is in the stretch state.

[0060] After the secondary processor circuit **120** determines that a prying action occurs, in step **818**, the secondary processor circuit **120** instructs the primary processor circuit **110** or the headset **100** to execute a predefined function corresponding to the prying action.

[0061] The embodiment of FIG. 8 highlights at least the following advantages. When different users wear the headset **100**, the headset **100** adaptively update the threshold values for determining the usage status according to different head sizes, and further adjust third threshold value Q3 for determining the prying action accordingly. The embodiment allows the headset **100** to be controlled by different users with customizable prying actions, while false action sensing in the secondary processor circuit **120** and the sensor circuit **130** are effectively reduced.

[0062] The possible functions derived from various measured value changes or prying actions will be illustrated in FIG. 9.

[0063] FIG. 9 is a flowchart of the handset control method according to a further embodiment of the present application. The embodiment discusses a state machine diagram of the headset **100** switching between four states based on the usage status. Based on the operation principle introduced in FIGS. 2 to 8, further applications for various states are shown in FIG. 9. The four states include the open state, the non-wearing state, the wearing state, and the power off state. The corresponding processes are presented in different partitions in FIG. 9.

[0064] Given that the process starts from step **900**, wherein the headset **100** operates in the open state. As shown in FIG. 3, the open state can be a state where the first earphone part **310** and the second earphone part **320** are apart by open state distance **DO**. The headset **100** may operate as, but not limited to, a desk speaker playing music without putting on the user head **400**. In step **902**, the primary processor circuit **110** of the headset **100** controls the wireless communication circuit **150**, the first audio module **210** and the second audio module **220** to play music or answer a call. The music played may be an audio file read from the memory device **140**. In actual operation, the headset **100** also comprises a first audio circuit **142** and the second audio circuit **144** coupled to the primary processor circuit **110**, configured respectively to play first channel audio and second channel audio under the control of the primary processor circuit **110**. The headset **100** also comprises an audio capture circuit **146**, coupled to the primary processor circuit **110**, configured to capture voice input. When the headset **100** is in the open state, the primary processor circuit **110** controls the wireless communication circuit **150**, the first audio circuit **142**, the second audio circuit **144** and the audio capture circuit **146** to play music or make voice calls.

[0065] It can be understood that while the steps are processed, the secondary processor circuit **120** and the sensor circuit **130** continue to monitor the sensor data and calculate the measured value in the background. To further optimize the power efficiency, the secondary processor circuit **120** may also enter the sleep mode when certain conditions are met, and is designed to be awakened by the sensor circuit **130**. For example, the sensor circuit **130** may transmit a wake-up signal to the secondary processor circuit **120** when a sensed voltage exceeds a predetermined offset.

[0066] In step **904**, if the secondary processor circuit **120** detects through the measured value of the sensor circuit **130** that a distance **D** between the first earphone part **310** and the second earphone part **320** changes to non-wearing distance **DN** and remains stable around non-wearing distance **DN**, it is determined that the headset **100** is in a non-wearing state, and therefore step **910** is proceeded. On the other hand, if step **904** is not true, the workflow continues to step **906**.

[0067] In step **906**, the secondary processor circuit **120** checks if the headset **100** has not been played for a long time. If yes, the secondary processor circuit **120** actively notifies the headset **100** to power off in step **930**. In practical operation, the headset **100** may also include a timer module **170**, coupled to the primary processor circuit **110** and the secondary processor circuit **120** to count a time value. The timer module **170** may be implemented by software code executable in the secondary processor circuit **120**, or by an independent hardware circuit. When the primary processor circuit **110** is not performing an audio play operation, the secondary processor circuit **120** monitors the time value in the timer module **170**. When the secondary processor circuit **120** determines that the primary processor circuit **110** does not play sound for more than a predetermined time limit, for example, ten minutes, then the secondary processor circuit **120** indicates that the headset **100** is powered off.

[0068] When the headset **100** is switched to a non-wearing state, the process starts from step **910**. In step **910**, the headset **100** operates in the non-wearing state. In one embodiment, the primary processor circuit **110** may actively stop playing music and enter the sleep mode as soon as the headset **100** is switched to the non-wearing state. Meanwhile, only the secondary processor circuit **120** and the sensor circuit **130** maintain active to keep the sensor data monitored. In step **912**, the secondary processor circuit **120** checks whether the wearing state is detected. If so, step **920** is

processed. If not, step **914** is processed. The principle and process of detecting usage status are shown in steps **800** to **806** of FIG. **8**, which will not be repeated.

[0069] In step **914**, the secondary processor circuit **120** also checks whether the open state is detected. In other words, when the distance  $D$  between the first earphone part **310** and the second earphone part **320** is changed from non-wearing distance  $DN$  to open state distance  $DO$  and remains stable around open state distance  $DO$  for a period of time, the secondary processor circuit **120** determines that an open state is detected, and therefore processes step **900**. If step **914** does not hold, the workflow continues with step **916**.

[0070] Step **916** is analogous to step **906**, wherein the secondary processor circuit **120** checks whether the headset **100** has not been played for a long time. If so, the workflow of the headset **100** proceeds to step **930**.

[0071] When the headset **100** is switched to the wearing state, the process starts from step **920**. In step **920**, the headset **100** operates in the wearing state. In one embodiment, the primary processor circuit **110** in the wearing state is awoken from the sleep mode, so as to run various functions in real time, such as voice call, music playback, voice recording, and etc. Meanwhile, in step **922**, the secondary processor circuit **120** detects whether the headset **100** is removed from the user head **400**. If the secondary processor circuit **120** detects the removal action, in addition to the headset **100** switching to the non-wearing state in step **910**, the primary processor circuit **110** also stops current music play, call or recording operation in step **924** to enter the sleep mode. The determination of the removal action, for example, comprises a secondary processor circuit **120** determining the distance  $D$  between the first earphone part **310** and the second earphone part **320** through the sensor data of the sensor circuit **130**. When the distance  $D$  is changed from wearing state distance  $DW$  to non-wearing distance  $DN$ , and stays stable around non-wearing distance  $DN$  for a period, for example, more than three seconds, it is determined that the headset **100** has entered the non-wearing state.

[0072] If step **922** is negative, the headset **100** continues to step **926**, and the secondary processor circuit **120** cooperates with the sensor circuit **130** to detect whether the headband **202** is stretched. The detailed principle of detecting the prying action is described in steps **810** to **818** of FIG. **8** and therefore is not repeated hereafter. If it is determined in step **926** that a prying action is detected, the primary processor circuit **110** proceeds to step **928** to pause the music play or mute an ongoing voice call. Furthermore, the function corresponding to the prying action may be dependent with ongoing operations. If the music is paused, the prying action un-pauses the music play. If a call rings, the prying action is to pick up the call. Furthermore, a consecutive number of prying actions may be associated to divergent functions, such as music fast turn, previous song, next song, and etc. In summary, based on the sensor data and measured value provided by the secondary processor circuit **120** and the sensor circuit **130**, the corresponding running functions can be flexibly defined.

[0073] If step **928** is negative, then the headset **100** continues with step **929**. Step **929** is analogous to step **916** and step **906**. It is determined by the secondary processor circuit **120** whether the headset **100** has not been played for a long time. If so, the headset **100** proceeds with step **930**.

[0074] In step **930**, when the headset **100** is in the power off state, most circuits in the headset **100** stop consuming power. Conventionally, a physical switch is required to turn on a powered off device. The embodiment, on the other hand, utilizes the existed component the sensor circuit **130** to achieve the effect of automatic powering up. The secondary processor circuit **120** and the sensor circuit **130** are further configured to operate continuously with low power consumption. When there is any structural deformation or tension variation on the headband **202**, it is inferred that someone may want to use the headset **100**. Meanwhile, the secondary processor circuit **120** determines whether the collected sensor data meets a wake-up condition in step **932**. If so, the headset **100** is woke up to operate in the non-wearing state. The wake-up condition may comprise the cases where the user actions are detected, such as opening, wearing, and removal of the headset **100**.

[0075] In a further derived embodiment, the threshold values used to determine various states may be configured through software pre-installed on a host device, such as a mobile phone app, which is then stored in the memory device **140** through wireless communication. When the secondary processor circuit **120** determines the usage status, the threshold values are read from the memory device **140**. Alternatively, the secondary processor circuit **120** updates the threshold values, and writes the updated threshold values into the memory device **140**.

[0076] In a further derived embodiment, the headband **202** of the headset **100** may be an elastic material for providing wearable fixation. The headband **202** can be in the form of a strap or a bracket. After the sensor circuit **130** is combined with the headband **202**, the sensor data can be generated according to the elastic deformation state or tension change on the headband **202**. As before, the sensor data can be a waveform signal formed by the change of voltage or current, or a digital signal, such as a digital value incremented with a step value 0.34 mv.

[0077] The embodiment of the present application highlights at least the following advantages. The sensor circuit **130** of the headset **100** can detect a variety of usage statuses, enabling a variety of customizable control functions. In addition, the basic monitoring function of the secondary processor circuit **120** enables the primary processor circuit **110** to enter the sleep mode when no sound is played, so as to save more power. Furthermore, the threshold value for determining the usage status can be dynamically adjusted to meet unique needs from various users. Overall, the embodiment of the application enables the headset **100** to play more functions and improve efficiency.

[0078] In further derived embodiments, the primary processor circuit **110** and the secondary processor circuit **120** can be implemented in the same chip, logically divided into primary and secondary processors, and powered separately. The sensor circuit **130** mentioned in the embodiment may be a pressure sensor, but can also be a deformation sensor to detect the deformation on the headband **202**. The sample rate for the sensor circuit **130** to collect the sensor data can be once every 10 milliseconds (MS).

[0079] It is to be understood that the term “comprises”, “comprising”, or any other variants thereof, is intended to encompass a non-exclusive inclusion, such that a process, method, article, or device of a series of elements not only include those elements but also comprises other elements that are not explicitly listed, or elements that are inherent to such a process, method, article, or device. An element defined by the phrase “comprising a . . . ” does not exclude the presence of the same element in the process, method, article, or device that comprises the element.

[0080] Although the present invention has been explained in relation to its preferred embodiment, it does not intend to limit the present invention. It will be apparent to those skilled in the art having regard to this present invention that other modifications of the exemplary embodiments beyond those embodiments specifically described here may be made without departing from the spirit of the invention. Accordingly, such modifications are considered within the scope of the invention as limited solely by the appended claims.

## Claims

1. A headset for audio play, comprising: a sensor circuit, arranged to operably generate sensor data based on structural deformations of the headset; and a control module, coupled to the sensor circuit, comprising: a secondary processor circuit, arranged to operably receive the sensor data and calculate a measured value from the structural deformations of the headset; wherein the secondary processor circuit is arranged to operably determine a usage status of the headset based on the measured value and one or more threshold values; wherein the secondary processor circuit is arranged to adaptively adjust at least one of the one or more threshold values based on the usage status.

2. The headset as claimed in claim 1, wherein the control module further comprises a primary

processor circuit; when the secondary processor circuit determines from the usage status that the headset requires the primary processor circuit to function, the secondary processor circuit issues an event notification to the primary processor circuit, causing the primary processor circuit to be awakened from a sleep mode and to control the headset in response to the usage status.

**3.** The headset as claimed in claim 1, wherein: when the measured value is lower than a first threshold value, the secondary processor circuit determines that the usage status is a non-wearing state; when the measured value is greater than or equal to the first threshold value, and lower than a second threshold value, the secondary processor circuit determines that the usage status is an open state; when the measured value is greater than or equal to the second threshold value, and lower than a third threshold value, the secondary processor circuit determines that the usage status is a wearing state; and when the measured value is greater than or equal to the third threshold value, the secondary processor circuit determines that the usage status is a stretch state.

**4.** The headset as claimed in claim 3, wherein: the secondary processor circuit is arranged to operably monitor the measured value; and when the usage status is the non-wearing state and the secondary processor circuit detects that the measured value is increased to a first value greater than the first threshold value and maintains substantially around the first value for a time period, the secondary processor circuit determines that the usage status is switched to the wearing state, and configures the second threshold value to be the first value subtracting a predetermined offset.

**5.** The headset as claimed in claim 4, wherein: the secondary processor circuit configures the third threshold value to be the second threshold value plus a fixed number; and when the headset operates in the wearing state, and the secondary processor circuit detects that the measured value is increased to exceed the third threshold value, the secondary processor circuit determines that the usage status is the stretch state.

**6.** The headset as claimed in claim 4, wherein: when the headset is in the wearing state, and the secondary processor circuit detects that the measured value is increased to exceed a predetermined value within a default time, the secondary processor circuit determines that the usage status is in the stretch state.

**7.** The headset as claimed in claim 3, wherein: the control module further comprises a primary processor circuit; when the headset is in the wearing state, and the secondary processor circuit detects based on the measured value that the usage status is changed to the non-wearing state, the secondary processor circuit notifies the primary processor circuit to stop an audio play operation, and notifies the primary processor circuit to enter a sleep mode.

**8.** The headset as claimed in claim 3, wherein: the control module further comprises a primary processor circuit; when the headset operates in the wearing state, and the secondary processor circuit detects based on the measured value that the usage status is changed to the stretch state, the secondary processor circuit notifies the primary processor circuit to stop an audio play operation, or notifies the primary processor circuit to mute a call.

**9.** The headset as claimed in claim 1, wherein when the secondary processor circuit receives the sensor data, the secondary processor circuit performs a jitter filtering on the sensor data, and cumulates or averages the jitter filtered sensor data collected within a first duration, to generate the measured value.

**10.** The headset as claimed in claim 1, further comprising: a wireless communication circuit, arranged to perform a wireless communication with a host device; and a memory device, adaptable for storing data and at least one of the one or more threshold values; wherein: the secondary processor circuit utilizes the wireless communication circuit to receive from the host device configurations related to the at least one of the one or more threshold values, and to store the configurations in the memory device.

**11.** The headset as claimed in claim 1, further comprising: a headband, coupled to the sensor circuit and the control module, arranged to be resilient and providing fixation of the headset; wherein the sensor circuit detects deformations of the headband to generate the sensor data; and the sensor data

comprises voltage values.

**12.** A headset control method, for controlling a headset comprising a sensor circuit and a control module, wherein the control module comprises a secondary processor circuit coupled to the sensor circuit; the headset control method comprises: utilizing the sensor circuit to detect structural deformations of the headset to generate sensor data; utilizing the secondary processor circuit to receive the sensor data, and to calculate a measured value from the structural deformations of the headset; the secondary processor circuit determines a usage status of the headset based on the measured value and one or more threshold value; and the secondary processor circuit adaptively adjusts at least one of the one or more threshold values based on the usage status.

**13.** The headset control method as claimed in claim 12, further comprising: when the secondary processor circuit determines from the usage status that the headset requires a primary processor circuit to function, utilizing the secondary processor circuit to issue an event notification to the primary processor circuit, causing the primary processor circuit to be awakened from a sleep mode and to control the headset in response to the usage status.

**14.** The headset control method as claimed in claim 12, further comprising: when the measured value is lower than a first threshold value, the secondary processor circuit deeming that the usage status is a non-wearing state; when the measured value is greater than or equal to the first threshold value, and lower than a second threshold value, the secondary processor circuit deeming that the usage status is an open state; when the measured value is greater than or equal to the second threshold value, and lower than a third threshold value, the secondary processor circuit deeming that the usage status is a wearing state; and when the measured value is greater than or equal to the third threshold value, the secondary processor circuit deeming that the usage status is a stretch state.

**15.** The headset control method as claimed in claim 14, further comprising: the secondary processor circuit monitoring the measured value; and when the usage status is the non-wearing state, and the secondary processor circuit detects that the measured value is increased to a first value greater than the first threshold value and maintains substantially around the first value for a time period, the secondary processor circuit deeming that the usage status is switched to the wearing state, and configuring the second threshold value to be the first value subtracting a predetermined offset.

**16.** The headset control method as claimed in claim 15, further comprising: the secondary processor circuit configures the third threshold value to be the second threshold value plus a fixed number; and when the headset operates in the wearing state, and the secondary processor circuit detects that the measured value is increased to exceed the third threshold value, the secondary processor circuit deeming that the usage status is the stretch state.

**17.** The headset control method as claimed in claim 15, further comprising: when the headset is in the wearing state, and the secondary processor circuit detects that the measured value is increased to exceed a predetermined value within a default time, the secondary processor circuit deeming that the usage status is in the stretch state.

**18.** The headset control method as claimed in claim 14, further comprising: when the headset is in the wearing state, and the secondary processor circuit detects based on the measured value that the usage status is changed to the non-wearing state, the secondary processor circuit notifying a primary processor circuit to stop an audio play operation, and notifying the primary processor circuit to enter a sleep mode.

**19.** The headset control method as claimed in claim 14, further comprising: when the headset operates in the wearing state, and the secondary processor circuit detects based on the measured value that the usage status is changed to the stretch state, the secondary processor circuit notifying a primary processor circuit to stop an audio play operation, or notifying the primary processor circuit to mute a call.

**20.** The headset control method as claimed in claim 12, further comprising: wherein when the

secondary processor circuit receives the sensor data, the secondary processor circuit performing a jitter filtering on the sensor data, and cumulating or averaging the jitter filtered sensor data collected within a first duration, to generate the measured value.

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